

ARNOB PROTIM ROY

+88 01772624924

pr.arnob@gmail.com

<https://trewto.github.io>

EDUCATION

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

M.Sc. in Transportation Engineering in Department of Civil Engineering

CGPA 3.83/4.00 MOI: English, Passing Year: Sep 2026

Postgraduate Fellowship Awardee, granted for academic excellence

Thesis: GENERATIVE ARTIFICIAL INTELLIGENCE BASED MODELLING OF PEDESTRIAN GROUP CROSSING BEHAVIOUR AT INTERSECTIONS IN DHAKA CITY (Supervisor: Dr. Md. Hadiuzzaman)

Developed a computer vision and machine learning framework for pedestrian crossing-intention modelling and trajectory forecasting. Evaluated multiple classifiers for intention prediction and LSTM/CVAE-based approaches for trajectory prediction using individual, social, demographic, and traffic-context information.

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

B.Sc in Civil Engineering (Major in Transportation Engineering, Minor in Geotechnical Engineering)

CGPA 3.36/4.00, MOI: English, Passing Year: July 2024

Thesis: A STUDY OF THE CIRCULAR WATERWAY AROUND DHAKA CITY (Supervisor: Dr. Hasib Mohammed Ahsan)

ADVANCED CROSS-DEPARTMENT GRADUATE COURSEWORK

CSE 6709 - Deep Learning (CSE, BUET)

CSE 6701 - Neural Networks (CSE, BUET)

CSE 6510 - Natural Language Processing (CSE, BUET)

CSE 6514 - Big Data Analytics (CSE, BUET)

EEE 6608 - Machine Learning and Pattern Recognition (EEE, BUET)

RESEARCH INTERESTS

Computer vision for transportation analytics

Spatiotemporal prediction and forecasting of traffic and mobility patterns

Data-driven modeling of transportation behavior and safety

PUBLICATIONS

Peer-Reviewed Conference Papers

Arnob Protim Roy, Md. Zunaid Farouque, Bharga Narayan Tarafder, and Nasrin Jahan Shila. (2026). Machine learning-based travel cost prediction for rickshaw trips in Dhaka city. In Proceedings of the 8th International Conference on Civil Engineering for Sustainable Development (ICCESD 2026), Bangladesh.

Arnob Protim Roy and Hasib Mohammed Ahsan. (2024). SWOT analysis of the circular waterway around Dhaka city. In Proceedings of the 7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024), Bangladesh.

Manuscripts Under Review

Arnob Protim Roy and Annesha Enam. (2026). Computer vision-based traffic-state estimation for heterogeneous traffic from CCTV recordings. Manuscript submitted to the International Conference on Civil Engineering 2026 (ICCE 2026).

Anika Mowrin, Nazmul Haque, **Arnob Protim Roy**, and Md. Hadiuzzaman. (2026). Trajectory-based pedestrian crash risk estimation using extreme value theory. Manuscript submitted to the International Conference on Civil Engineering 2026 (ICCE 2026).

Paromita Datta Ray and **Arnob Protim Roy**. (2026). Machine learning-based prediction of unconfined compressive strength of sand-tire derived aggregate mixtures using cross-study literature data. Manuscript submitted to the International Conference on Civil Engineering 2026 (ICCE 2026).

PROJECTS

- November 2025 Traffic Survey & Analysis - Cumilla City Bound Corridor**
Conducted analysis of a 7-day video-based classified traffic survey on the Dhaka-Chattogram (N1) corridor, evaluating ADT, peak-hour metrics, PCU conversion, and safety conflicts to support TIA and geometric design
- September 2025 Traffic Impact Analysis Report: Two Storied (G+1) Residential Building, Satarkul, Dhaka**
Applied TIA techniques using PTV VISSIM with focus on residential-scale traffic and to support Rajdhani Unnayan Kartripakkha (RAJUK) approval process.
- March 2025 Traffic Impact Analysis Report: 10-Story Multipurpose Building in Manda, Dhaka**
Analyzed trip generation, capacity, and future traffic scenarios using HCM methodology and PTV VISSIM; developed mitigation strategies for Dhaka Transport Coordination Authority(DTCA) approval.
- July 2023 to June 2024 Capstone Project: Redevelopment of Mouchak Market as a Modern Shopping Complex**
Collaborated in a six-member team to redevelop Mouchak Market in Dhaka with sustainable civil engineering solutions. Conducted structural and traffic analyses while preparing cost estimation, scheduling, and resource allocation plans aligned with urban development standards.

TECHNICAL PROJECTS

- Traffic Speed Estimation using Camera-GPS Calibration
- AI-Assisted Image Annotation Tool with Human-in-the-Loop Workflow
- Face-based Reverse Search Engine using Computer Vision and DBMS
- Vehicle Detection, Tracking and counting on road Using Computer Vision
- License Plate Number Detection using OCR and Deep Learning
- Shortest Path Detection Program from Map/Image Using Advanced Graph Theory
- Weather Prediction Model in Machine Learning Using Bangladeshi Weather Data
- Pygame Based Bacteria Evolution Simulation
- Borno CMS : Content Management System Web App (PHP and MySQL Based)
- Web-based Quiz Management System using Laravel MVC Framework

TECHNICAL SKILLS

- **Artificial Intelligence:** Machine Learning, Deep learning, Specialized in Computer Vision (YOLO)
- **Data Science & Numerical Analysis:** Excel
- **Design & Modelling Software:** AutoCAD, SAP2000, ETABS, MATLAB, PTV VISSIM
- **Programming:** PHP, Python, C++, C, Javascript, R
- **Web Development:** HTML, CSS, PHP, MySQL, Laravel
- **CMS & Server Management:** WordPress, Joomla, cPanel, phpMyAdmin, XAMPP/LAMP Stack
- **Additional Tools:** n8n, Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Docker

FIELD VISIT

- Representing BUET in the “Tarunner Startup and Sombvobonar Bangladesh”- program by ICT Ministry (2026)
- A visit to Bangladesh Military Stadium with msc supervisor for field inspection (2026)
- A visit to Dhaka Circular Waterway, with BIWTA, BIWTC, DTCA & BUET Team (2026)
- A visit to Police officer housing society assisting preliminary roundabout design inspection (2026)
- A one day visit to Sayedabad Water Treatment Plant in Level 4, Term 2 from BUET (2024)
- Dhaka Airport Third Terminal Construction Site course in Level 4, Term 2 from BUET (2024)

AWARDS AND ACHIEVEMENTS

- Postgraduate Fellowship, BUET - Awarded for academic excellence in M.Sc

- National & Regional Winner at National High School Programming Contest 2015, 2017

Workshop

- An in-depth workshop on protein folding and AlphaFold, Hosted by BRAC University (2026)

VOLUNTEERING EXPERIENCES

Bangladesh Climate Science Olympiad 2021 *Coordinator (2021)*

- Coordinated the national program ensuring smooth execution, logistics, and participant engagement.
- Organized seminars, training camps, and question preparation/review for national winners.

International Astronomical Union *Volunteer (2019 - 2020)*

- Coordinated various national programs, seminars, and workshops promoting astronomy.
- Contributed to the Bangla translation of IAU astronomy booklets, notably Jotibidda Moddakotha, enhancing accessibility for Bangla-speaking audiences.
- Led the 12th Abdul Jabbar Astronomy Workshop at BUET with IAU and NARIT, featuring lectures and hands-on sessions.

Naming Exoworld-Project Bangladesh *Coordinator, Bangladesh National Committee (2019-2020)*

- Led Bangladesh's participation in the IAU100 Global Event, enabling the naming of exoplanet HD 148427b as "Tondra" and its host star HD 148427 as "Timir".
- Managed logistics, academic support, and nationwide student engagement while liaising with IAU

LANGUAGE SKILLS

English (IELTS 6.5) , **Bangla,Hindi, Basic French** (Reading, Writing)